# Supplementary material to: A physics-informed digital twin for tube-life-aware operation of industrial steam methane reformers under uncertainty

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## S1. Digitised verification data

The verification targets of Section 3.1 are the axial profiles of Fig. 3 of Xu and Froment (1989, Part II). The figure was digitised curve by curve [TODO: name the tool (e.g. WebPlotDigitizer), the image resolution and the axis-calibration procedure] and stored as one tidy CSV per curve in `data/literature_validation/digitized/` with the columns `z_m`, value and a `source` annotation. Curves: methane conversion, $CO_2$ conversion, process-gas temperature, inner-wall temperature, outer-wall temperature and total pressure, over the heated length of 11.12 m; the remaining length to 12 m is treated as adiabatic. The outer-wall curve is interpolated by a shape-preserving cubic (PCHIP) and used as the boundary condition; the digitising uncertainty is estimated at [TODO: quote the estimated reading uncertainty in K and in conversion] from repeat readings of the axis ticks. The feed and geometry are those of Table 2 of the source, with a bed voidage of 0.526 and an equivalent ring diameter of 9.44 mm fitted to the pressure curve [TODO: confirm that fitting the particle diameter to the pressure drop is acceptable].

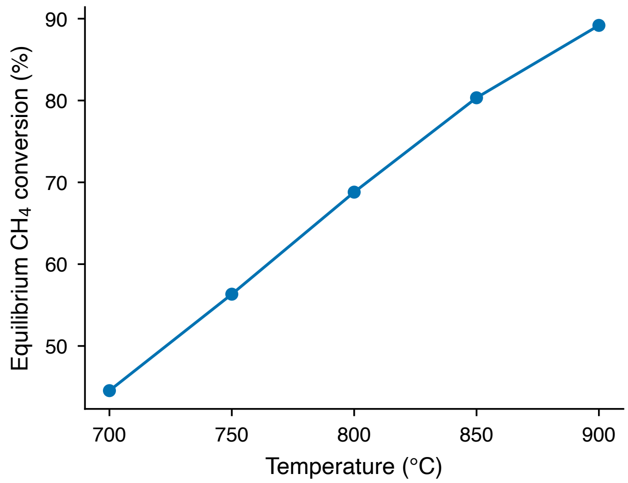


Figure S1: Equilibrium methane conversion at 25 bar and steam-to-carbon 3 from Cantera (GRI-Mech 3.0), used as a sanity check of the thermodynamic basis. Source: `notebooks/00_sanity_check.ipynb`.

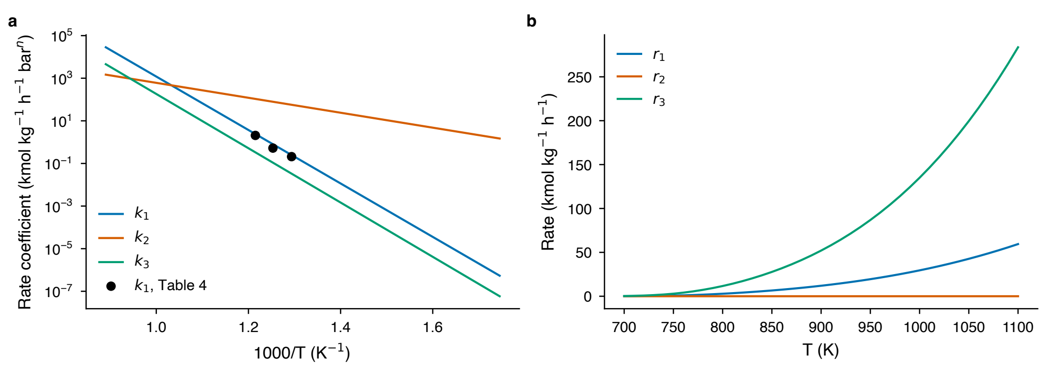


Figure S2: Xu–Froment kinetics as implemented: (a) rate coefficients $k_1$–$k_3$ against reciprocal temperature with the Table 4 values of the source for $k_1$, and (b) rates $r_1$–$r_3$ at 25 bar for a feed of $CH_4$:$H_2O$:$H_2$ = 1:3:0.1. Source: `data/kinetics/xu_froment_1989.yaml`.

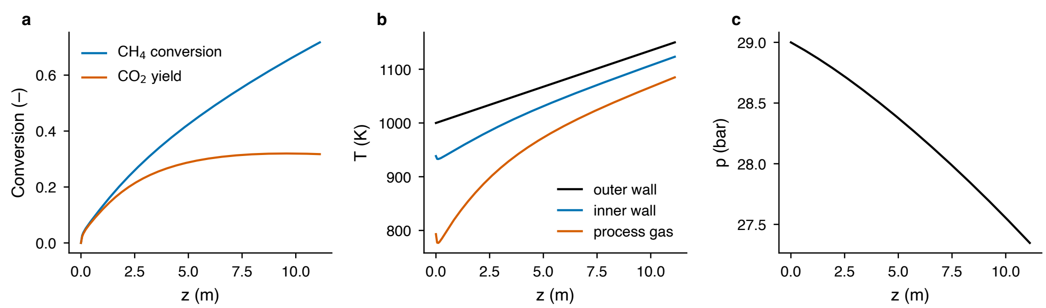


Figure S3: Stand-alone tube model with a linear outer-wall profile from 1000 to 1150 K on the Xu–Froment geometry: (a) conversion and $CO_2$ yield, (b) wall and gas temperatures, (c) pressure.

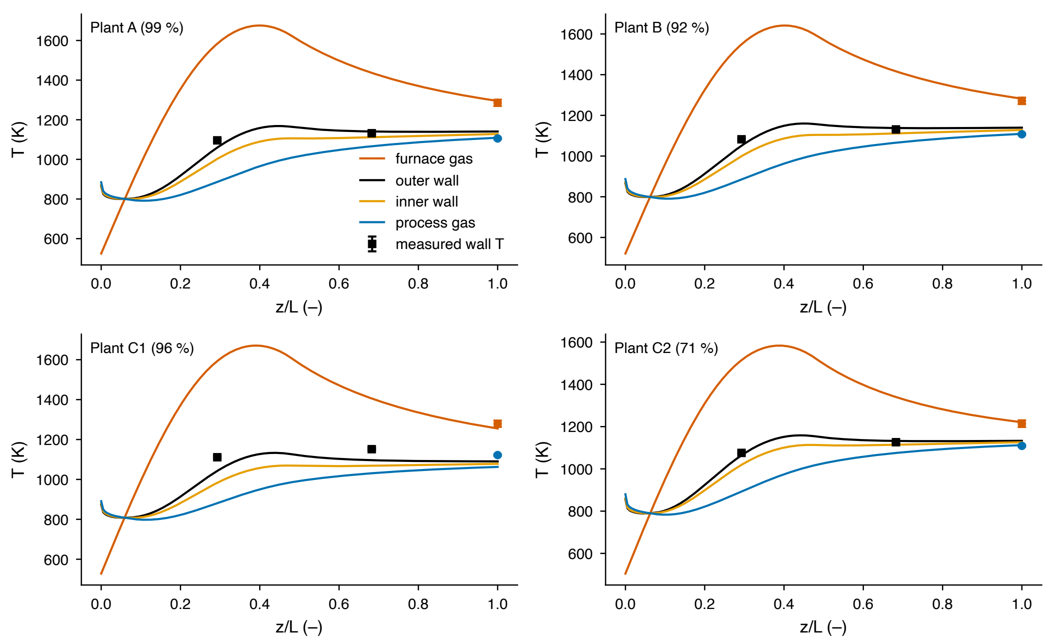


Figure S4: Coupled model with the literature (uncalibrated) parameters on all four Latham plant cases, including Plant C1 which is excluded from calibration. Source: `latham2008_plant_cases.csv`, `latham_fit.yaml` (baseline).

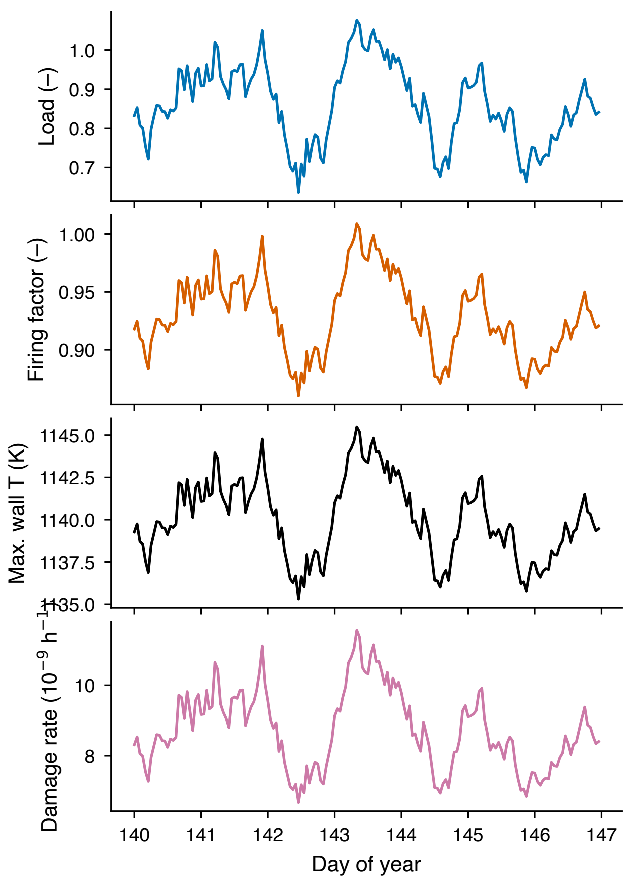


Figure S5: One week (days 140–147) of the renewable-following scenario S3: load, firing factor solved by the outlet-temperature controller, hot-spot temperature and hourly damage rate. Source: `data/scenarios/hourly_v2_S3_renewable.csv.gz`.

Table S1: Operating inputs and their ranges (v2 operating space); base = calibrated Plant A. Source: `data/design/parameter_ranges_v2.csv`.

| Input | Unit | Min | Max | Base | Justification (abridged) |
|---|---|---|---|---|---|
| steam-to-carbon | – | 2.5 | 4.0 | 2.89 | carbon-formation limit below ≈2.5; 4.0 upper for hydrogen plants |
| load (feed per tube) | – | 0.60 | 1.10 | 1.00 | turndown 50–60 %; 110 % short-term overload |
| specific firing factor | – | 0.85 | 1.15 | 1.00 | burner control range |
| inlet temperature | K | 800 | 900 | 884.6 | Plant A 611 °C; cracking in pre-heat coils above ≈920 K |
| inlet pressure | bar | 25 | 33 | 30.06 | industrial 20–40 bar |
| catalyst activity | – | 0.10 | 0.30 | 0.20 | Latham fitted 0.20 for a mid-campaign catalyst |
| excess air | % | 5 | 25 | 21.6 | upper bound raised from 20 to 25 % so that the calibrated Plant A value lies inside the space |

## S2. Additional figures

## S3. Operating space

## S4. Version note

All results in the main text are from pipeline version v2, which differs from the exploratory version v1 in the excess-air range (5–25 % instead of 5–20 %), in the use of the calibrated excess air as the base of the scenarios and the optimisation (instead of 20 %) and in the Monte Carlo sample size (2048 instead of 4000). A line-by-line comparison of the two versions is provided in `data/design/v1_vs_v2_diff.md`; the largest effect of the change is a base hot spot about 3 K cooler and hence a longer base life, which shifts every relative life metric by the same denominator.

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